

4.4. Consider the 11g sliding-IF receiver shown in Fig. 4.116.

- Determine the required LO frequency range.
- Determine the image frequency range.

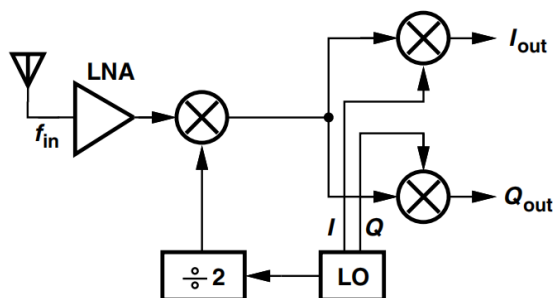


Figure 4.116 Sliding-IF RX for 11g.

4.6. The sliding-IF architecture shown in Fig. 4.117 is designed for the 11a band.

- Determine the image band.
- Determine the interferer frequencies that can appear in the output baseband as a result of mixing with the third harmonic of the first LO or the third harmonic of the second LO.

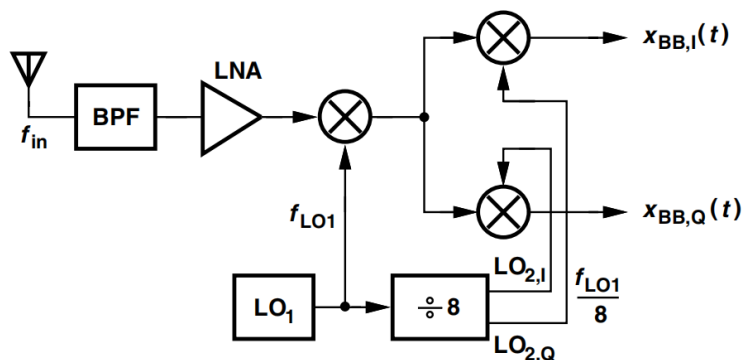


Figure 4.117 Sliding-IF RX for 11a.

4.9. In this problem we wish to study an interesting effect that arises from LO leakage in direct-conversion receivers [20]. Consider the LO leakage in a direct-conversion receiver, $V_0 \cos \omega_{LO} t$. Suppose this leakage is added to an amplitude-modulated interferer, $V_{int}(t) \cos \omega_{int} t$, and the result experiences third-order nonlinearity in the LNA (or downconversion mixer).

- Determine the components near the carrier at the LNA output.
- Determine the resulting baseband components and whether they corrupt the desired signal.

[20] M. Conta, private communication, Feb. 2011.

4.17. Prove that the IRR of the Hartley architecture is given by $(\omega_{IF}/\Delta\omega)^2$ at an intermediate frequency of $\omega_{IF} + \Delta\omega$ if $\omega_{IF} = (R_1 C_1)^{-1}$.

4.24. Assume a second-order parallel RLC tank is excited by a current source containing a component at ω_0 and another at $3\omega_0$. Prove that, if the tank resonates at $3\omega_0$, then the first harmonic is attenuated by approximately a factor of $8Q/3$ with respect to the third harmonic.